1. *Aluminum.* Aluminum is produced by adding large amounts of electricity to aluminum oxide. A manager of a smelting plant considering entering the aluminum industry estimates that combining e thousand kilowatt-hours of electricity and a metric tons of aluminum oxide creates:

$$q = f(e, a) = 10(9 * e * a)^{1/4}$$

metric tons of aluminum metal.

a) The price of electricity is \$50 per thousand kilowatt-hours and the price of aluminum oxide is currently \$200 per metric ton. Solve for the MP of e and a.

MP_e=______________________ MP_a=______________________

Solution:

$$\boldsymbol{MP_e = 10(0.25)(9 * e * a)^{-3/4} 9a}$$

$$\boldsymbol{MP_a = 10(0.25)(9 * e * a)^{-3/4} 9e}$$

b) In a cost-minimizing production plan, how many thousand kilowatt-hours of electricity should be used for every metric ton of aluminum oxide?

KW Hours (000's) = ____________________________________

Solution:

We solve for this by equating the ratio of the marginal products to the price ratio:

$$\boldsymbol{\frac{MP_e}{MP_a} = \frac{p_e}{p_a}}$$

$$\boldsymbol{\frac{10(0.25)(9 * e * a)^{-3/4} 9a}{10(0.25)(9 * e * a)^{-3/4} 9e} = \frac{50}{200}}$$

$$\boldsymbol{\frac{a}{e} = \frac{1}{4}}$$

implying that $\boldsymbol{e^* = 4a^*}$.

c) Suppose the smelting plant manager wants to produce 3,000 tons of aluminum. How many thousand kilowatt-hours of electricity and how many tons of aluminum oxide should be used?

KW Hours of Electricity (000's) = ______________________

Tons of Aluminum Oxide = __________________________

Solution:

We solve this by using the production function and the cost-minimizing input ratio found in part (a).

$$\mathbf{3000 = 10(9 * e * a)^{1/4}}$$

$$\mathbf{= 10(9 * 4a * a)^{1/4}}$$

After rearranging to solve for o:

$$\sqrt{\boldsymbol{a}} = \frac{\mathbf{300}}{\mathbf{36^{1/4}}}$$

$$\boldsymbol{a}^* = \frac{\mathbf{300^2}}{\sqrt{\mathbf{36}}} = \mathbf{15,000}$$

$$\boldsymbol{e}^* = \mathbf{4(15,000) = 60,000}$$

Given the units, this is 60,000,000 kilowatt-hours.

d) Once the manager enters the aluminum industry, she realizes that a smelting plant producing q metric tons of aluminum actually has a daily variable cost of:

$$VC(q) = 0.5q^2$$

Throughout, let p denote the price per metric ton of aluminum. Note that you do not need to use the production function from (a) to answer the following.

What is the plant's short-run supply curve, q(p)? Also, assuming there are 100 such plants (with an identical cost structure) in the world, what is the global supply of aluminum $Q_S^G(p)$?

Short-Run Supply, q(p): ____

Global Supply, $Q_S^G(p)$?: ____

Solution:

To get the short-run supply, we first need to solve for marginal cost:

$$\boldsymbol{mc = \frac{1}{2}(2)q = q}$$

Setting $\boldsymbol{mc = p}$, we can see that the short-run supply is given by $\boldsymbol{q = p}$. Therefore,

$$\boldsymbol{Q_S^G(p) = 100p}$$

e) Daily demand for aluminum in the global market is:

$$P = \frac{8{,}000}{3} - \frac{1}{300} Q_D^G(P)$$

What is the equilibrium price and daily quantity of aluminum sold per plant?

Equilibrium Price: ________________________

Quantity Sold Per Plant: ____________________

Solution:

$$\boldsymbol{Q_G^S = Q_G^D}$$

$$\boldsymbol{100p = 800{,}000 - 300P}$$

$$\boldsymbol{P^* = 2{,}000}$$

$$\boldsymbol{Q^* = 100P^* = 100(2000) = 200{,}000}$$

Since there are 100 plants, q=2,000.

f) Now consider the entry of Iceland into the global aluminum market. Suppose that there are 10 aluminum producers in Iceland that can produce relatively cheaply using geothermal energy, with:

$$VC^I(q) = 0.3q^2$$

How much would each of these plants want to supply at the price solved for in (d)?

Quantity: ______________________________

Solution:

Solving for one the supply function for one of the producers in Iceland yields:

$$\boldsymbol{q^I(P) = \frac{5}{3}P}$$

So, they would want to supply q=3,333.

g) Of course, we would expect their entry to lower prices. Solve for the new global supply curve and equilibrium price of aluminum.

Global Supply, $Q_S^G(p)$?: ______________________________________

Equilibrium Price: ______________________

Which type of plant produces more aluminum – a traditional one, or an Icelandic one?

- ☐ Traditional
- ☐ Icelandic

Global supply is given by:

$$\boldsymbol{\widetilde{Q_G^S} = 100P + 10\left(\frac{5}{3}P\right) = \frac{350}{3}P}$$

After equating the market demand and the new global supply, we can solve for the equilibrium price as before, yielding: $\widetilde{P^*} = \$1,920$. The market price has therefore declined by \$80 due to the new producers.

It is trivial that the more efficient firms (Icelandic ones) produce more than traditional ones. You can confirm this by plugging the price into each plant's supply curve.

2. *Picking up the Crumbs.* Crumbs Bakery has determined their daily total cost function (where quantity is in dozens of cupcakes) is

$$TC(q) = 225 + 6q + q^2$$

Unfortunately for Crumbs, cupcakes are commodities and there is free entry, making the market perfectly competitive.

(a) (3 points). What is Crumbs' supply function for dozens of cupcakes (for the range of prices where they are willing to supply any), **in terms of the market price for a dozen cupcakes**?

Supply function: $Q_S(P) = \frac{1}{2}P - 3$____

Answer

Given the cost function, Crumbs' marginal cost is

$$MC(q) = \frac{dTC(q)}{dq} = 6 + 2q$$

since Crumbs is in a perfectly competitive market, the will be price takers and set price equal to marginal cost

$$P = 6 + 2q$$

since this expression relates the number of dozens of cupcakes they will supply and the price thy receive, we can solve this for q to get their supply function

$$q = -3 + \frac{1}{2}P$$

(b) (3 points). If the current prevailing market price in the short run is \$4 per cupcake (so \$48 per dozen), how many cupcakes will Crumbs supply, and what will their daily profit be?

Supplied Quantity (dozens): $Q_S(P) = 21$______________________________________

Daily Profit: $\$216$____

Answer

Given their supply function, at the prevailing market price of \$48 they will supply

$$Q_S(P) = -3 + \frac{1}{2} \times 48 = 21$$

and their daily profits are

$$\pi = P \times Q - TC(Q) = P \times Q - 255 - 6 \times Q - Q^2$$

$$\pi = 48 \times 21 - 225 - 6 \times 21 - (21)^2 = \$216$$

(c) (3 points). Fads come and go, and the cupcake boom appears to be turning into a cupcake bust: market prices for cupcakes are falling. In the short-run, the fixed cost component of total costs is unavoidable. What is the price (per dozen cupcakes) below which Crumbs would stop baking entirely in the short term?

Price \$6____

Answer

Given that Crumbs has fixed costs that are unavoidable in the short run, they will still produce (even if they make negative profits) as long as

$$\min(AVC) < P$$

The average variable cost curve is

$$AVC = \frac{TVC}{q} = 6 + q$$

this cost curve is minimized when q is as small as it can be, near zero (technically, the average variable cost curve is not defined at $q = 0$). For simplicity, just ignore this technicality so that $\min(AVC) = 6$. If you make the assumption that Crumbs can only produce discrete numbers of dozens of cupcakes, then $q = 1$ is as small as it can be and $\min(AVC) = 7$. So if Crumbs receives a price below \$6 (\$7) they will stop baking entirely in the short run.

(d) (4 points). In the long run, there is free entry, and indeed several competitors enter the cupcake market with the same cost function as Crumbs. What will be the long-run equilibrium price for a dozen cupcakes?

Price $36____

Answer

In the long run, the output decision is the same as in the short run, so Crumb's supply function doesn't change. However, since there is free entry and positive profits to be made, firms will enter and exit until the long-run equilibrium price is

$$P = \min(ATC)$$

The average total cost is

$$ATC = \frac{TC(q)}{q} = \frac{225}{q} + 6 + q$$

to find the value of q where the ATC is minimized, take the derivative of ATC with respect to q and solve the first order condition for q:

$$\frac{\partial ATC}{\partial q} = -\frac{225}{q^2} + 1 = 0$$

$$q^2 = 225 \rightarrow q^* = 15$$

so plugging $q^* = 15$ into the ATC function will tell us the minimal value of the ATC:

$$\min(ATC) = \frac{225}{q^*} + 6 + q^* = \frac{225}{15} + 6 + 15 = 36$$

so $P = \min(ATC) = 36$ is the long-run equilibrium price.